ADITYA CAKTI CHANDRASA

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Education

Universitas Padjadjaran Sumedang, Indonesia
Bachelor of Engineering – Electrical Engineering. **GPA: 3.56 / 4.00** Expected Graduation: 2028
Relevant Coursework: Electrical Circuit Analysis, Microcontroller Systems, Control Systems, Electromagnetic Fields.

Professional Experience

Freelance Web Developer Bandung, Indonesia
Futura 2026 Mar 2026 – Present

- Built and deployed a full-stack platform (Next.js, React, TypeScript, Supabase) that handles registration, scheduling, and payments for 200–300 participants.
- Designed the authentication and database layer, with Zod and React Hook Form for input validation.

PT PLN (Persero) ULTG Bandung Barat Bandung, Indonesia
Power Transmission Engineer Intern Jul 2026 – Aug 2026

- Ran thermographic inspections on 150 kV transmission towers to find thermal anomalies on jumper clamps and repair sleeves.
- Calculated temperature differentials against a 32°C ambient baseline to classify severity and prioritize maintenance.
- Found a critical overheating fault at Tower 22 (35.18°C differential) and a warning at Tower 37 (18.00°C), and recommended immediate corrective action.
- Traced the anomalies to mechanical vibration loosening and hydraulic crimping failures.

Universitas Padjadjaran Sumedang, Indonesia
Programming Laboratory Assistant Feb 2026 – Jun 2026

- Mentored 7 engineering students in programming fundamentals, algorithm design, and logic formulation.
- Wrote and ran weekly assessments on data structures and code implementation, and helped debug logic errors during lab sessions.

Technical & Academic Projects

Photovoltaic Microgrid Management (Academic Research) Sumedang, Indonesia
Academic Project Aug 2026 – Present

- Authored a paper, "Pengelolaan Jaringan Mikro Photovoltaic oleh Badan Usaha Milik Desa untuk Ketahanan Pangan Terisolasi", on off-grid solar power for isolated food storage and agricultural facilities.
- Proposed a management model where Village-Owned Enterprises (BUMDes) handle long-term maintenance, local administration, and financial viability.

Risk Profiler: API-Based Alternative Risk Analysis System Bandung, Indonesia
Project Manager Feb 2026 – Jul 2026

- Led a 4-person team building a credit risk profiling platform for MSMEs using alternative data, for the PIDI Digdaya Hackathon BI 2026.
- Designed the logic that turns non-traditional data into credit scores, and the backend architecture behind it.

MOTOSENSE: Vibration Monitoring System Sumedang, Indonesia
Academic Project Feb 2026 – Jun 2026

- Built an IoT vibration monitor for motorcycle CVT units using a Raspberry Pi 3B and ADXL345 sensor, with local alerts (LED, buzzer) and an online mode that syncs to InfluxDB.
- Deployed a web dashboard showing continuous vibration data and long-term degradation trends.

Spatial Metal Field Display Sumedang, Indonesia
Academic Project May 2026 – Jun 2026

- Built an ESP32 metal detector using an LC resonance circuit and hardware Pulse Counter (PCNT), with a 35-40 turn search coil at about 157 kHz.
- Applied EMA filtering to reduce the effect of parasitic capacitance on the readings.
- Verified Ampere's and Lenz's laws by measuring shifts in effective inductance from sheet currents and ferromagnetic cores.

Organizational & Leadership Experience

Himpunan Mahasiswa Teknik Elektro (HMTE) UNPAD Sumedang, Indonesia
Information and Communication Jan 2025 – Present

- **Head of Information and Communication** *(Jan 2026 – Present)*: Leading IT and media strategy, managing a cross-functional team across two sub-departments. Directed digital transformation from manual administrative processes to automated data management systems. Led a strategic shift from visual-heavy designs to narrative-driven copywriting.
- **Staff of Relations Department** *(Jan 2025 – Dec 2025)*: Executed external communication strategies and managed stakeholder relationships. Managed "University Explore" (UNEX), facilitating strategic comparative study visits to 3 top-tier universities, surpassing the initial target by 50%.

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Project Director Feb 2026 – May 2026

- Oversaw the project manager committee and delivered the event on a 6 million IDR budget.

Electro Future Talk 2026 Sumedang, Indonesia
Talkshow Moderator May 2026

- Moderated a 2-hour panel with 3 multinational corporate alumni.

Futura National Seminar & Robotics Competition Sumedang, Indonesia
Co-Project Manager Feb 2025 – Nov 2025

- Co-directed planning for a 9-month national robotics competition and technology seminar.
- Grew team participation by 60% (33 to 53 teams) and launched the Transporter and Sumo categories together for the first time since 2017.
- Managed 50+ staff across committees, 150+ participants on the field, and the event budget.

Certifications, Training & Competitions

- **Competitions:** Project Manager, Risk Profiler, PIDI Digdaya x Hackathon BI 2026.
- **Technical Training:** AWS AI Academy 2026; Dicoding Certifications (Software Development, Machine Learning)
- **Language Proficiency:** TOEFL ITP (Score: 607/677) – Advanced Proficiency (Mar 2024); EF SET English Certificate (Score: 79/100) – C2 Proficient (Feb 2025).

Technical Skills

- **Hardware:** ESP32, Arduino, Raspberry Pi, NI Multisim.
- **Software:** C/C++, Python, TypeScript, JavaScript, Next.js, React, Supabase
- **Design:** Figma, Canva